HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Authored precise, step-by-step technical solutions and "golden responses" for complex electrical engineering problems, including circuit analysis and signal processing, serving as benchmarks for AI model training.
- Evaluated AI-generated code and mathematical proofs against engineering standards such as IEEE, identifying logical errors and refining model reasoning on core electrical engineering concepts.
- Designed advanced electrical engineering problems across domains like control systems and power electronics, specifically to rigorously test and assess AI model performance.
- Developed Python scripts to automate data extraction and transformation for training datasets, processing approximately 50,000 records weekly for AI model evaluation.
- Contributed to the maintenance and enhancement of enterprise Java/.NET applications, ensuring operational stability through proactive monitoring and incident response.
- Wrote and optimized SQL queries for data validation and ad-hoc reporting, managing schemas and data integrity across PostgreSQL and MySQL databases.
- Containerized microservices using Docker and deployed them to AWS EC2 instances, configuring basic resource limits and networking for predictable performance.

SKILLS

Core Engineering – Electrical Engineering, Circuit Analysis, Power Systems, Signal Processing, Electromagnetics, Control Systems, Power Electronics, Embedded Systems

AI & Machine Learning – AI Model Training, Data Annotation, Data Quality Assessment, AI Reasoning Refinement, Technical Accuracy Evaluation

Programming Languages – Python, Java, SQL, R

Databases – PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn

Methodologies – Agile, Scrum

PROJECTS

AI Model Training for Electrical Engineering Concepts

- Authored detailed, ground-truth solutions for advanced electrical engineering problems, covering topics like VLSI design and telecommunications, to provide accurate training data for AI models.
- Performed rigorous technical accuracy audits on AI-generated circuit diagrams and mathematical derivations, providing structured feedback to improve model understanding and output quality.
- Created a Python-based data pipeline to extract, clean, and format engineering problem sets and solutions, enabling efficient ingestion into AI training frameworks.

Automated Technical Documentation Review System

- Developed a Python script to parse and analyze technical documentation and AI-generated responses, flagging inconsistencies with engineering standards (e.g., IEEE).
- Implemented logic to identify and report on logical fallacies or factual errors in AI-generated reasoning for complex control system scenarios.
- Designed an output format for the review system that clearly highlights discrepancies and provides actionable feedback for AI model refinement.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)